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SUPERSEDING
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FEDERAL SPECIFICATION

COPPER RODS AND SHAPES; AND FLAT PRODUCTS WITH FINISHED EDGES (FLAT WIRE, STRIPS AND BARS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers copper rods, shapes, and flat products with finished edges (flat wire, strip, and bar) used for general fabricating purposes, but does not include bus bar for electrical applications nor strips and bars furnished with slit, sheared, sawed, or machined edges (see 6.3).

1.2 Classification.

1.2.1 *Forms and tempers.* The material shall be furnished in the following forms and tempers, as specified (see 6.2):

Forms:

Rods	
Shapes	
Bars	Flat products with finished edges
Strips	
Flat wire	

Tempers:
Soft
Hard

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 102 — Preservation, Packaging and Packing Levels.
Fed. Std. No. 146 — Tolerances For Copper And Copper Base Alloy Mill Products.
Fed. Test Method Std. No. 151 — Metals; Test Methods.
Fed. Std. No. 185 — Identification Marking of Copper and Copper Base Alloy Mill Products.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington 25, D. C.

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FSC 9525, 9530, 9535, and 9540

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Military Specification:

MIL-P-3993 — Preparation of Shipment and Storage of Copper and Copper Alloy Mill Products.

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military specifications, standards, in connection with specific procurement functions drawings and publications required by contractors should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Manufacture.** Copper rods, shapes, flat wire, finished-edge strip, and bars shall be manufactured either by hot rolling or by extrusion and shall be finished by such cold-rolling or cold-drawing, including straightening, as may be required to produce material conforming to the requirements of this specification. A finish anneal shall be given to material ordered in the soft condition.

3.2 **Chemical composition.** Unless otherwise specified (see 3.4 and 6.1), material shall contain not less than 99.88 percent copper, small amounts of silver being counted as copper.

3.3 **Mechanical properties.**

3.3.1 *Tensile and hardness properties.* Rods, shapes, and bars shall meet the tensile requirements shown in table I. Flat wire and finished-edge strip shall meet the tensile and hardness requirements shown in table II.

TABLE I. — *Tensile requirements for rods, shapes and bars*

Form and temper	Size (diameter or distance between parallel faces)	Tensile strength minimum	Elongation in 2 inches minimum[1]
	Inches	*P.s.i.*	*Present*
Rods and bars, soft	All sizes	[2]37,000	25
Rods, hard	Up to 0.250, incl.	50,000	—
	Over 0.250 to 0.375, incl.	45,000	10
	Over 0.375 to 1, incl.	40,000	12
	Over 1.000 to 2, incl.	[2]35,000	15
	Over 2.000 to 3, incl.	33,000	15
Bars, hard	Over 0.188 to 0.375, incl.	42,000	10
	Over 0.375 to 0.500, incl.	40,000	12
	Over 0.500 to 2, incl.	33,000	15
	Over 2.000 to 4, incl.	32,000	15
Shapes, soft	All sizes	[2]35,000	25
Shapes, hard	All sizes	32,000	15

[1]When round specimens other than type R1 of method 211 of Fed. Test Method Std. No. 151 are used, elongation shall be determined on a gage length of four times the diameter except that the minimum gage length shall be 1 inch.

[2]Maximum.

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TABLE II. — *Tensile and hardness requirements for flat wire and finished -edge strip*

Temper	Thickness	Tensile strength		Elongation in 2 inches, minimum	Rockwell hardness, F scale (60 kg. load, 1/16 inch ball)	
		Minimum	Maximum		In rolls, on bucks, reels or spools	In straight lengths
	Inch	*P.s.i.*	*P.s.i.*	*Percent*		
Soft	Up to 0.010, incl.	———	———	20	———	———
	Over 0.010 to 0.020, incl.	———	40,000	25	———	———
	Over 0.020 to 0.050, incl.	———	[1]38,000	[1]25	55 max.	62 max.
	Over 0.050 to 0.188, incl.	———	[1]37,000	[1]25	55 max.	62 max.
Hard	Up to 0.020, incl.	43,000	58,000	—	———	———
	Over 0.020 to 0.125, incl.	[1]43,000	———	[1]10	85 to 97	85 to 97
	Over 0.125 to 0.188, incl.	[1]43,000	———	[1]12	80 to 95	80 to 95

[1]Required only when specified in the contract or order.

3.3.2 *Bending properties.* When specified (see 6.2), rods, flat wire, finished-edge strip, and bars in hard temper shall withstand being bent at room temperature through an angle of 120 degrees and in soft temper through an angle of 180 degrees around a radius equal to the minimum cross-sectional dimension of the specimen, and this dimension shall be radial to the bend without cracking on the outside of the bent portion.

3.4 Oxygen-free copper.

3.4.1. When oxygen-free copper is specified (see 6.2) the material shall contain not less than 99.95 percent copper (small amounts of silver being counted as copper) and shall be produced without the use of metallic or metalloidal deoxidizers.

3.4.2 *Resistivity.* For annealed oxygen-free copper, the electric resistivity, at a temperature of 20° C. (68° F.) shall not exceed 0.15329 ohm-gram/meter[2]. The density of copper shall be taken as 8.89 grams per cubic centimeter at 20° C.

3.4.3 *Freedom from cuprous oxide.* The oxygen-free copper shall be free from cuprous oxide, as determined by the test specified in 4.8.2.

3.4.4 *Embrittlement.* When specified, samples of oxygen-free copper shall not show gassing or open-grain structure characteristic of embrittlement when tested as in 4.8.3.

3.5 Tolerance. Tolerance shall not exceed that specified in the following sections of Fed. Std. No. 146:

Flat Products, with Finished Edges, Cold Drawn or Cold Rolled:
- Thickness Tolerance (Copper alloys)
- Width Tolerance (Copper alloys)
- Length Tolerance (Copper alloys)
- Schedule of Lengths (Copper alloys)
- Straightness Tolerances (Copper alloys)
- Standard Edge Contours (Copper alloys)

Rod, Cold Drawn:
- Diameter Tolerance (Copper alloys)
- Straightness Tolerances (Copper alloys)

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Length Tolerances (Copper alloys)
Schedule of Lengths (Copper alloys)

Rod, Hot Rolled, Round:
Diameter Tolerance (Copper alloys)

Bar, as Extruded:
Tolerances on Diameter or Distance Between Parallel Surfaces (Copper alloys)

Shapes:
Dimensional Tolerance (Copper alloys)
Length Tolerance (Copper alloys)
Straightness Tolerances (Copper Alloys).

3.5.1 Unless otherwise specified (see 6.2), flat wire, finished edge strip, and bar shall be furnished with either square or rounded corners at the option of the manufacturer.

3.6 **Identification marking.** Unless otherwise specified (see 6.2), marking shall be in accordance with Fed. Std. No. 185.

3.7 **Workmanship.** Flat wire, finished edge strip, bars, rods, and shapes shall be uniform in quality and temper and shall be sound, clean, smooth and free from injurious defects, including foreign material, pipes, slivers, laps, cracks, seams, scale, burrs, buckles, and damaged ends, corners, or edges.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facility and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirement.

4.1.1 *Lot.* Unless otherwise specified in the contract or order, a lot shall consist of 10,000 pounds, or fraction thereof, of material of the same form, temper, and size submitted for inspection at one time.

4.2 **Sampling.** Samples taken for purposes of the tests prescribed in this specification shall be selected in a manner to represent correctly the material furnished.

4.2.1 *Samples for chemical analysis.* From different representative pieces in each lot the number of samples for chemical analysis shall be selected as specified in table III. From each sample, not less than 2 ounces of clean millings, drillings, or clippings shall be obtained, placed in separate containers, and forwarded to a laboratory satisfactory to the bureau or agency concerned for analysis. A single analysis of a composite sample may be made.

TABLE III. — *Sampling for chemical analysis*

Pounds of material in lot	Number of samples[1]
Up to 5,000	2
5,001 to 10,000	4

[1]If the number of original pieces from which the material is processed is less than the number of samples specified, not more than one sample need be taken from each piece.

4.3 **Samples for mechanical tests.** Two tension, two hardness, and one bend test specimen shall be taken from each lot and each shall be selected from different pieces.

4.4 **Samples for other tests.** For oxygen-free copper, one resistivity test specimen, one microscope test specimen, and when specified, one embrittlement test specimen shall be taken from each lot, but each need not be selected from different pieces.

4.5 **Sampling for visual and dimensional examination.** If the weight of each piece is

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more than 150 pounds, every piece shall be examined. If the weight of each piece is less than 150 pounds, sample pieces shall be selected as specified in 4.5.1 and 4.5.2.

4.5.1 *Visual examination.* From each lot of material containing pieces weighing 150 pounds and under, a representative sample of material shall be selected in accordance with MIL-STD-105 at inspection level III with an AQL of 1.5 percent defective and shall be visually examined as specified in 4.6.1.

4.5.2 *Dimensional examination.* From each lot of material containing pieces weighing 150 pounds and under, a representative sample of material shall be selected in accordance with MIL-STD-105 at inspection level III with an AQL of 1.5 percent defective and shall be measured as specified in 4.6.2. The samples selected for dimensional examination may be the same as those selected for visual examination (see 4.5.1).

4.5.3 *Finished edge strip and flat wire.* When finished edge strip and flat wire are furnished in rolls, reels, or bucks, the portion for examination shall be taken from within ten feet of the outer end.

4.6 **Visual and dimensional examination.**

4.6.1 *Visual.* Each of the sample pieces selected in accordance with 4.5.1 shall be visually examined to determine compliance with this specification.

4.6.2 *Dimensional.* Each of the sample pieces selected in accordance with 4.5.2 shall be in compliance with the specified dimensions.

4.6.3 *Straightness.* Straightness shall be determined by placing the piece on a level surface so that the arc or departure from straightness is horizontal. The maximum depth of arc shall be measured to the nearest 1/32 inch by means of a straightedge and a steel scale.

4.6.4 *Inspection for preparation for delivery.* The preservation, packaging, packing and marking shall be examined to verify conformance with the requirements of section 5 of this specification.

4.6.5. **Rejection.** Any nonconforming piece in each examination sample shall be rejected, and if the number of nonconforming pieces in the sample exceeds the acceptance number specified in MIL-STD-105 of that sample, the entire lot shall be rejected, subject to the reexamination provisions of section 13 of MIL-STD-105.

4.7 **Test procedures.**

4.7.1 *Chemical analysis.* Samples selected in accordance with 4.2.1 shall be analyzed in accordance with the wet chemical method 111 and the spectrographic method 112 of Fed. Test Method Std. No. 151 to determine conformance with 3.2. A single analysis of a composite sample may be made. In case of dispute, the analysis by the wet method shall be the basis for acceptance.

4.7.2 *Tension tests.* Each of the specimens selected in accordance with 4.3 shall be subjected, in the presence of the inspector, to the tension test as follows:

4.7.2.1 All bars, rods, flat wire, and shapes shall be tested in full size when practicable, and the elongation measured in a gage length of four times the diameter of round specimens and of four times the minimum distance across flats of specimens other than round, except that the gage length shall be not less than 2 inches. When a machined specimen becomes necessary, enough metal may be removed from the gage section to meet the limitations of the testing machine, or the specimen may be machined to the form and dimensions in accordance with type

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R1 of method 211 of Fed. Test Method Std. No. 151. When a type R1 specimen is used, the elongation shall be measured in a gage length of 1 inch.

4.7.2.2. For bars, rods, and shapes up to 1-1/2 inches in diameter or minimum thickness the axis of the test specimen shall coincide with the central axis of the piece. For bars, rods, and shapes 1-1/2 inches and over in diameter or minimum thickness, when a machined specimen is used, the axis shall be located midway between the center and the rolled or drawn surface of the piece.

4.7.2.3 For finished edge strip, the test specimen shall be machined to the form and dimensions of type F2 of method 211 of Fed. Test Method Std. No. 151. The longitudinal axis of the specimen shall be parallel to the direction of rolling, drawing, or extruding.

4.7.3 *Bending tests.*

4.7.3.1 When specified, bending test specimens shall be machined to the form specified in method 231 of Fed. Test Method Std. No. 151.

4.7.4 *Hardness test.* Fat wire and finished edge strip shall be subjected to a Rockwell hardness test in accordance with method 243 of Fed. Test Method Std. No. 151 to determine conformance with table II. At least three tests shall be made on each specimen.

4.8 Microscopic, resistivity, and embrittlements tests for oxygen-free copper.

4.8.1 *Resistivity test.* The test specimen for the resistivity test may be in the form of bars, rods, or shapes. It shall be of uniform cross-section throughout its length within plus and minus 0.75 percent of the cross-sectional area. Wherever possible, it shall be the full cross-section of the material it represents, if the full cross-section is such that the uniformity of the cross-section area can be accurately determined. The specimens shall be annealed at approximately 500° C. (932° F.) for 30 minutes. Resistivity shall be determined in accordance with method 611 of Fed. Test Method Std. No. 151.

4.8.2 *Microscopic test.* The sample selected as specified in 4.4 shall be examined under a microscope at 75 diameters magnification to determine conformance with 3.4.3.

4.8.3 *Embrittlement test.* The sample selected in accordance with 4.4 shall be heated to a temperature of 800° to 875° C. (1,472° to 1,607° F.) for a period of 20 minutes in a furnace containing an atmosphere of hydrogen. After removal from the furnace, the sample shall be polished, etched if desired, and examined under a microscope at a magnification of 75 to 200 diameters. The specimen shall not show gassing or the open-grain structure characteristic of embrittlement.

4.9 Rejection and retests.

4.9.1 *Rejection.* If the test specimen fails to meet any of the requirements of this specification, the lot represented by the specimen shall be rejected subject to the retest provisions of Fed. Test Method Std. No. 151.

5. PREPARATION FOR DELIVERY

(For civil agencies only, Fed. Std. No. 102 should be referred to for definitions of the various levels of packaging protection for supplies and equipment).

5.1 Preservation, packaging, packing and marking. Material shall be preserved and packaged level A or C as specified (see 6.2); packed level A, B or C as specified (see 6.2) and marked in accordance with MIL-P-3993. Preservative as specified in MIL-P-3993 shall not be required for level A preservation and packaging.

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6. NOTES

6.1 **Intended use.** This specification covers copper flat wire, strip, bars, rods, and shapes intended for general fabrication purposes. For special uses, the compositions shown in table IV are commercially available. If required, these particular compositions may be purchased under this specification by specifying chemical composition and other special requirements in the contract or order.

TABLE IV. — *Compositions of copper bars, rods, and shapes*

Type of copper	Copper[1]	Phosphorus	Arsenic	Silver
	Percent	*Percent*	*Percent*	*Percent*
Electrolytic tough pitch	99.90	———	———	———
Fire refined tough pitch[2]	99.88	———	0.012, max.	———
Phosphorized, with high residual P	99.90	0.015-0.04	———	———
Oxygen-free, no residual deoxidants	99.95	———	———	———
Silver bearing, tough pitch	99.90	———	———	[3]
Silver bearing, phosphorized	99.90	0.04, max.	———	[3]
Silver bearing, oxygen-free	99.90	———	———	[3]

[1]Minimum, including silver.
[2]Antimony, 0.003 percent, max.; selenium plus tellurium, 0.025 percent, max.; nickel, 0.05 percent, max.; bismuth, 0.003 percent, max.; lead, 0.004 percent, max.
[3]Desired minimum must be specified (normally between 6 and 10 ounces per ton).

6.2 **Ordering data.** Procurement documents should specify the following:

(*a*) Title, number, and date of this specifation.
(*b*) Forms and tempers required (see 1.2).
(*c*) Whether bending test is required (see 3.3.2).
(*d*) Whether oxygen-free copper is required (see 3.4.1).
(*e*) Corners required, if other than square or rounded are desired (see 3.5.1).
(*f*) Whether marking other than that required by Fed. Std. No. 185 is required (see 3.6).
(*g*) Whether packaging should be in accordance with level A or C and packing in accordance with level A, B or C (see 5.1).

6.3 **Related specifications.**

6.3.1 Copper bus bars, including bars, rods, and shapes, should be purchased under QQ-C-504.

6.3.2 This specification covers strip and bars having drawn, extruded, or rolled (not previously slit) edges. QQ-C-576 covers strip and bars having machined, sawed, sheared, or slit edges.

6.4 Since the mechanical properties obtainable in shapes in the hard temper depend on the size and intricacy of the shape, the requirements for each order should be agreed upon between the contractor and the bureau or agency concerned.

6.5 **Definition of forms.** Bar, rod, shape, flat wire and finished edge strip are defined in Fed. Std. No. 146.

6.6 The chemical and mechanical property requirements of this specification are similar to ASTM B 133-58.

6.7 **Transportation description.** Transpor-

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tation descriptions and minimum weights applicable to this commodity are:

Bars or Rods

Rail: Bars, drawn, extruded or rolled, copper. Carload minimum weight 30,000 pounds.

Motor: Bars or rods, drawn, extruded or rolled, copper. Motor volume minimum weight 36,000 pounds.

Shapes

Rail: Blanks (unfinished shapes), not otherwise indexed by name, copper. Carload minimum weight 30,000 pounds.

Motor: Blanks (unfinished shapes), not otherwise indexed, copper. Motor volume minimum weight 30,000 pounds.

Strips

Rail: Strip, not otherwise indexed by name, copper. Carload minimum weight 30,000 pounds.

Motor: Strip, not otherwise indexed, copper. Motor volume minimum weight 30,000 pounds.

Wire

Rail: Wire, plain, copper. Carload minimum weight 30,000 pounds.

Motor: Wire, plain, copper. Motor volume minimum weight 30,000 pounds.

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